

5 (**long**) A complex variable z satisfies

$$\frac{|z - i|}{|z + i|} = a$$

where a is a constant.

- (a) Derive an equation for the locus of z on the Argand diagram. [12]
- (b) Find the condition that a should satisfy such that z has valid solutions with positive values on the imaginary axis. [6]
- (c) For $a = 0.9$, determine the intersections of the locus with the imaginary axis. [6]
- (d) Sketch the locus of z on the Argand diagram for the value of a in part (c). [6]